



PA-9

T-302-HONN, Reykjavik University
Computer Science

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1 Modeling (45 points)

1. (Total points: 10) Why do we want to have low coupling between services?

Answer: Low coupling between services means that changes in one service have little or no impact on others. This allows each service to be changed independently without impacting the other services. It just improves overall system flexibility and low coupling also makes the system more resilient, since a single failing service does not bring the rest down, that is the goal at least.

2. (Total points: 10) Why do we want to have high cohesion within our services?

Answer: High cohesion means that everything inside a service works together to perform one well defined purpose. It also makes the service easier to understand. Services with high cohesion are more maintainable and reusable since their responsibilities are clear and most importantly are also isolated.

3. (Total points: 10) Why is Content Coupling a particularly problematic form of coupling?

Answer: Content coupling is when one service stretches itself over and accesses or modifies the internal data or logic of another service. This completely breaks the main idea of distributed systems which is to have isolated and independent systems because changes in one service can directly break another. It essentially makes independent deployment nearly impossible.

4. (Total points: 15) Explain how it is also possible to model services after:

- **Volatility:** Services can be modeled based on how often different parts of the system change. Components that change together should belong to the same service. This reduces the need to redeploy unrelated functionality when something changes.
- **Data & Security:** Some data is very sensitive, so we need to keep the sensitive data all in one service and have a high attention to security on that service.
- **Technology:** Services can also be separated based on the technology stack they use and what tech stack is the best for a specific task in the service. For example, one service might be written in Python for machine learning tasks while another is written in Java for business logic. This also doesn't force one stack across the whole system.

2 Communication Styles (55 points)

1. (Total points: 10) Explain Asynchronous communication.

Answer: Asynchronous communication means that services don't wait for an immediate response when sending a message. Instead, the sender continues its work while the receiver handles the message later on its own pace. This allows for more decoupled and scalable systems, as it reduces temporal coupling.

2. (Total points: 10) Explain how event-driven architecture is naturally asynchronous.

Answer: Event-driven architectures are naturally asynchronous because services communicate through events rather than direct calls. When the sender wants something to happen it publishes an event (often placed on a queue) and other services that care about it, i.e. services that have "subscribed to the event", can react in their own time, which is most often just taking an entry off the queue and processing it and repeat. The publisher does not know or care who receives it or when. This is exactly the definition of asynchronous.